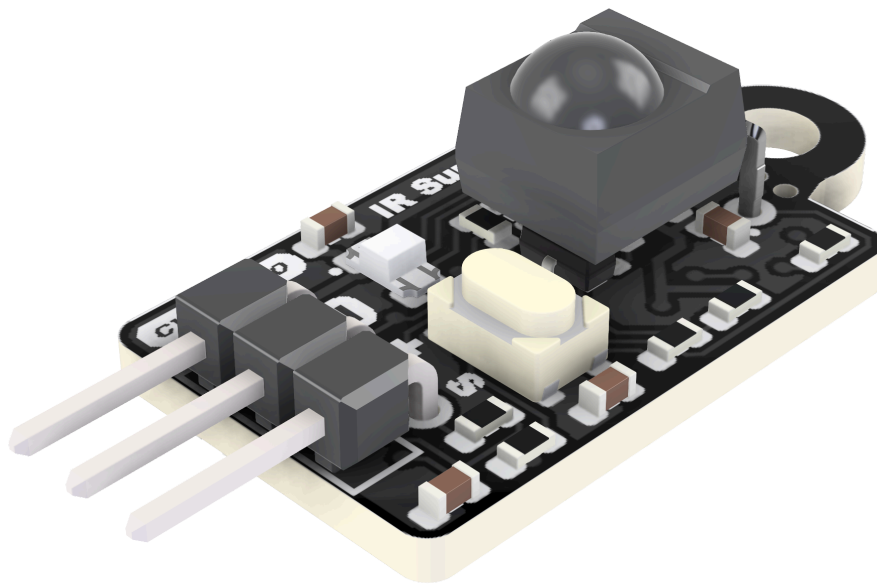




IR SUMO START



Datasheet

Rev 1.0
April 2025

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1. BOARD LAYOUT AND FUNCTION

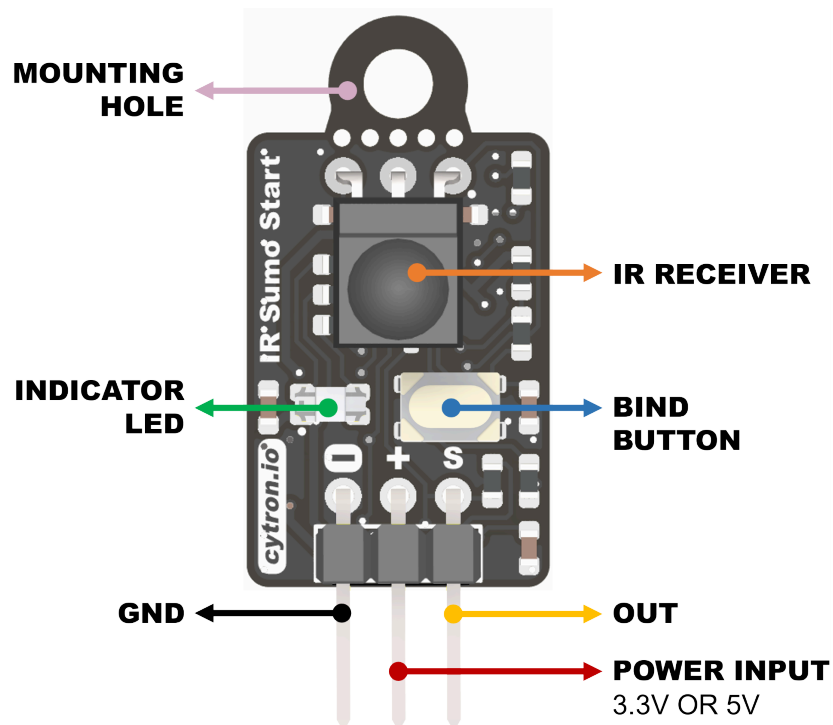


Figure 1: IR-SUMO-START Top View Layout Overview - Key Components

Function	Description												
IR Receiver	• Infrared receiver that receives IR signal from the remote/transmitter. • Built-in 38kHz bandpass filter.												
Indicator LED	• A RGB LED that used to indicate the active state/mode of the module. a. Startup Light up GREEN for 1 second, then turn off. b. Receive Mode The LED changes color and behavior to indicate the active state. Each state is activated by its corresponding IR command input. <table><tr><th>State</th><th>LED Behavior</th><th>Output Logic</th></tr><tr><td>Ready</td><td>Flashes PINK then turn to solid BLUE</td><td>LOW</td></tr><tr><td>Run</td><td>Solid GREEN</td><td>HIGH</td></tr><tr><td>Stop</td><td>Blinks RED slowly indefinitely</td><td>LOW</td></tr></table>	State	LED Behavior	Output Logic	Ready	Flashes PINK then turn to solid BLUE	LOW	Run	Solid GREEN	HIGH	Stop	Blinks RED slowly indefinitely	LOW
State	LED Behavior	Output Logic											
Ready	Flashes PINK then turn to solid BLUE	LOW											
Run	Solid GREEN	HIGH											
Stop	Blinks RED slowly indefinitely	LOW											

	c. Bind Mode In Bind Mode (activated by a long press of the bind button), the LED's behavior is as follows: 1. Initialization: The LED flashes RGB colors, then turns off, indicating the module is in Bind Mode. 2. IR input Binding: The module binds three IR input commands, one for each state in Receive Mode. The LED blinks in the corresponding color (blue for Ready, green for Run, red for stop) to confirm each successful binding.
Bind Button	• Push button used to enter bind mode. • Press and Hold the button until the Indicator LED flashes RGB colour.
Mounting Hole	• Breakout mounting hole allows for optional mounting.
Connector	• + : Power input, compatible with 3V3 and 5V • - : Ground (GND) • S : Output signal (0 or 1 Logic). The output voltage for HIGH logic is equal to power input voltage. ($V_s = V_{in}$)

Table 1: IR-SUMO-START Components and Their Functions

2. SPECIFICATION

No	Parameters		Min	Max	Unit
1	Power Input Voltage (+)		3.0	5.5	V
2	Output Signal Voltage*	Power Input (+) = 5V	0	5	V
		Power Input (+) = 3.3V	0	3.3	V
3	Output Signal Current	Power Input (+) = 5V	-	2.27	mA
		Power Input (+) = 3.3V	-	1.50	mA

Table 2: IR-SUMO-START Absolute Maximum Ratings

***Note:**

The output signal voltage (S) is equal to the power input voltage supplied. Do not exceed the maximum power input voltage limit.

3. DIMENSION

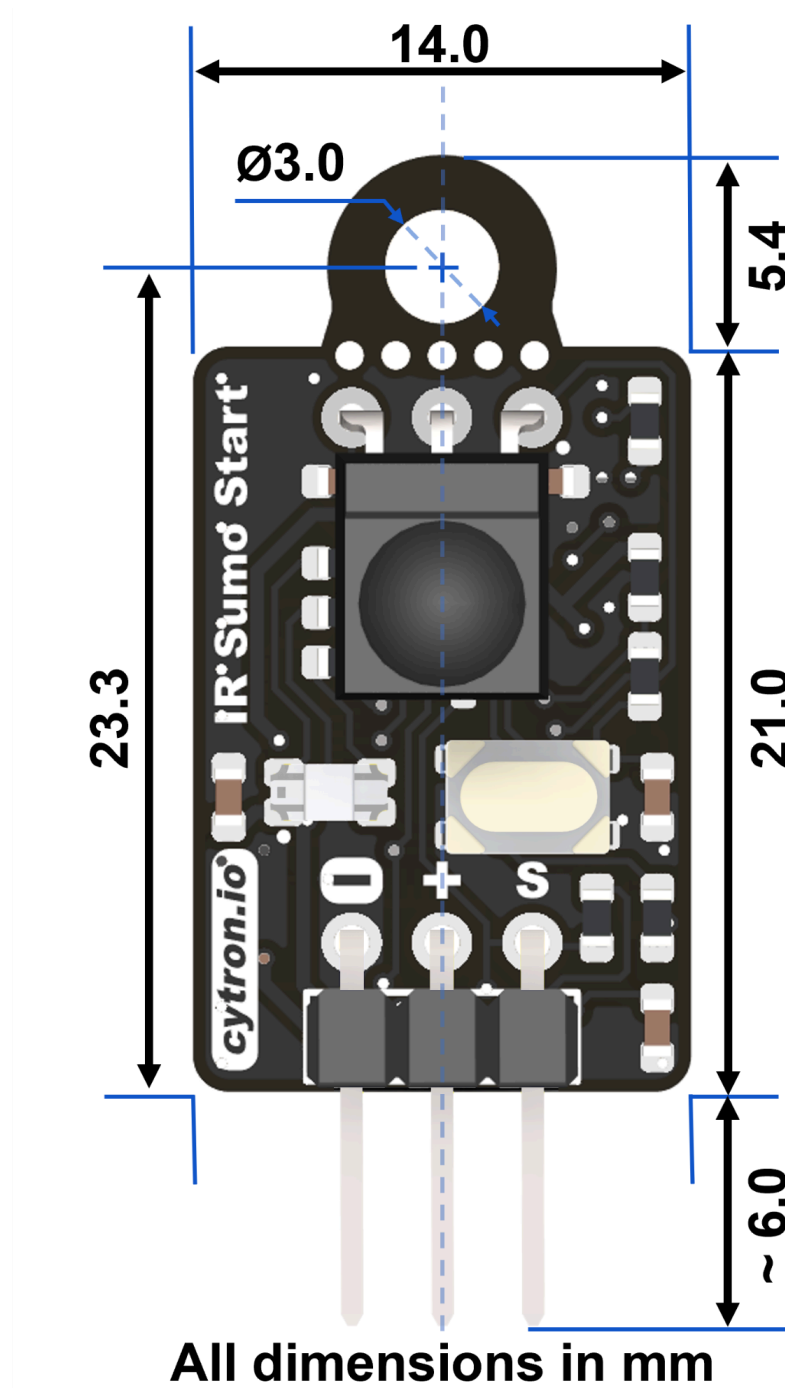


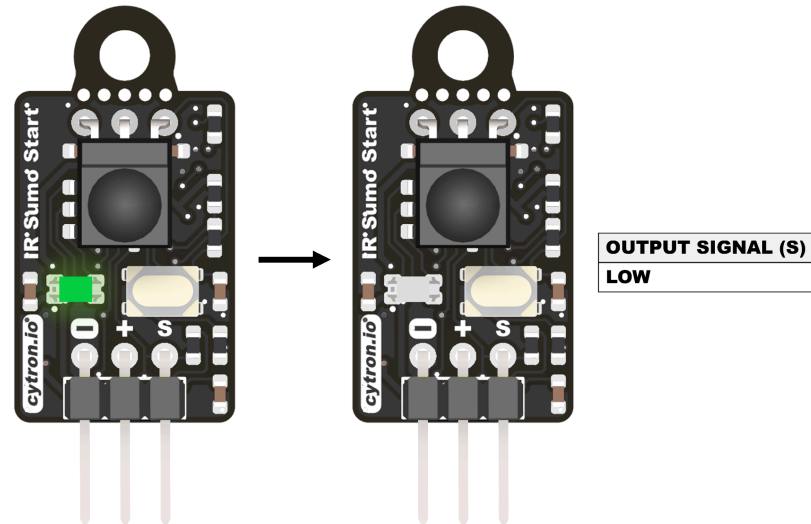
Figure 2: IR-SUMO-START Dimension (Top View)

→ [3D CAD files](#)

4. GETTING STARTED

a. On Start

The LED will light up **GREEN** for 1 second, then turn off. The module now is in **Receive Mode** and ready to respond to IR input. The output signal (S) state is LOW.

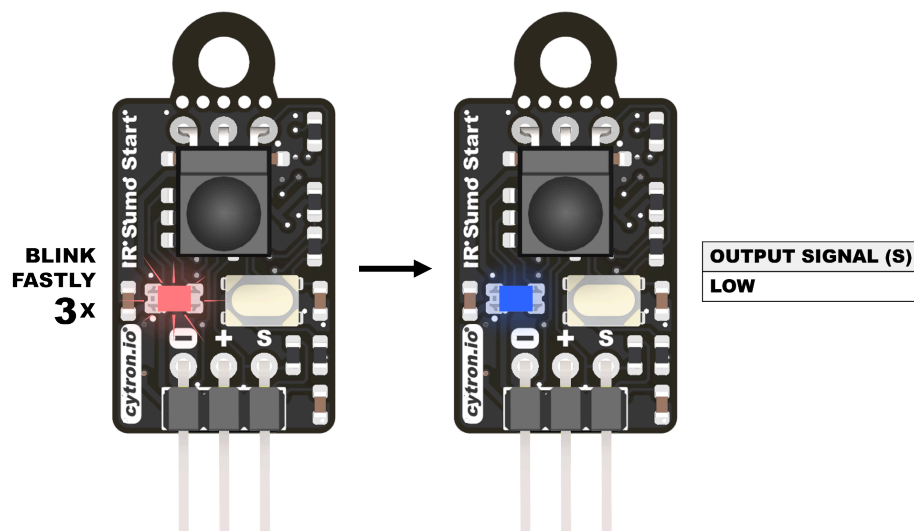


b. Receive mode

The module is ready to detect and respond to Infrared (IR) signals from any IR remote. It will react to the pre-bind IR command for each state (READY, RUN, STOP) and act accordingly;

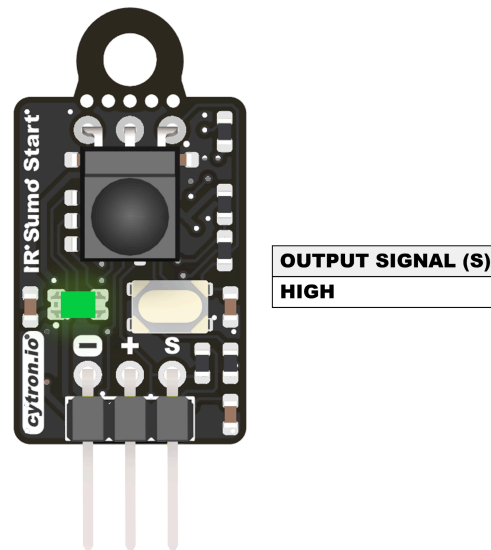
READY

The LED will blink **PINK** fastly, then change to static **BLUE**. The output signal (S) state stays LOW.



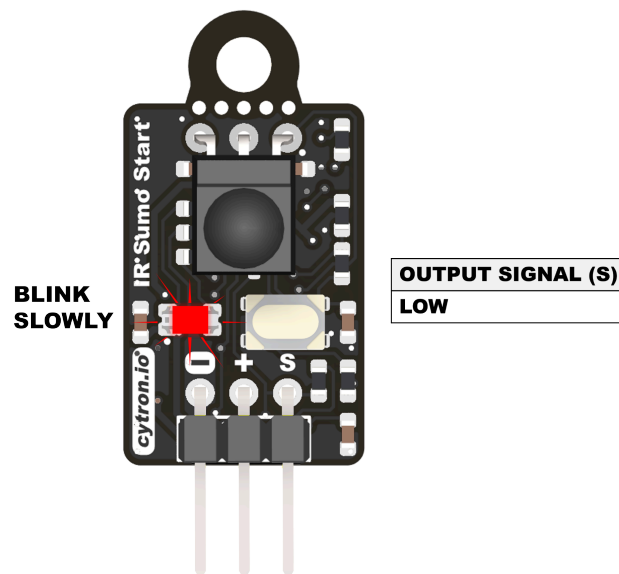
RUN

The LED will change to static **GREEN** and the output signal (S) state changes to HIGH. This should trigger the sumo robot to start.



STOP

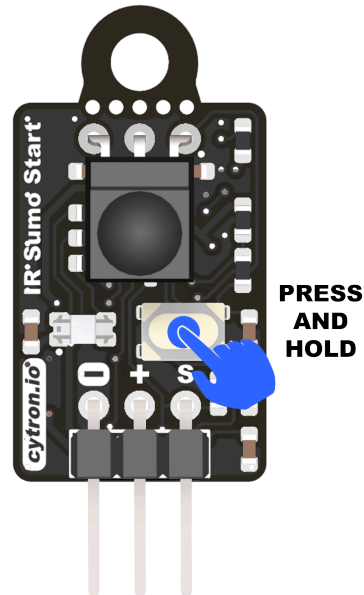
The LED will blink **RED** slowly indefinitely and the output signal (S) state changes to LOW. This should stop the robot immediately. The module needs to be manually reset to be used again for the next round.



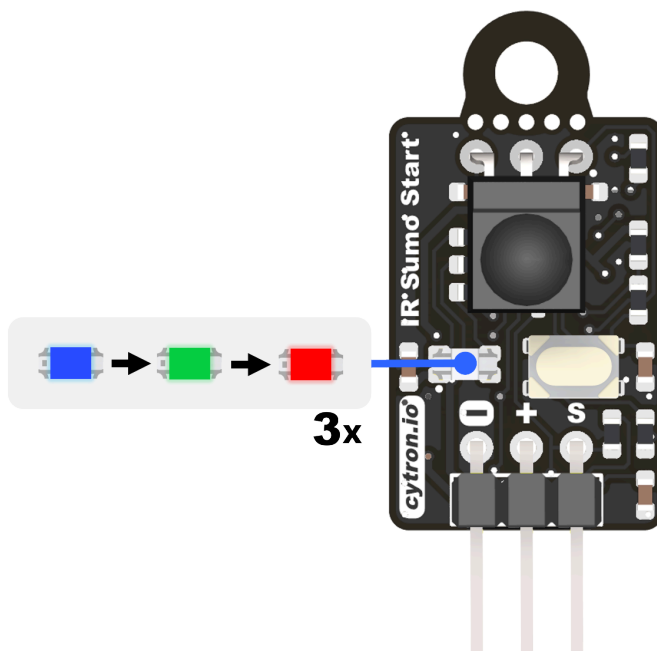
c. Bind mode

Entering bind mode:

1. Press and hold the bind button for more than 1 Second.

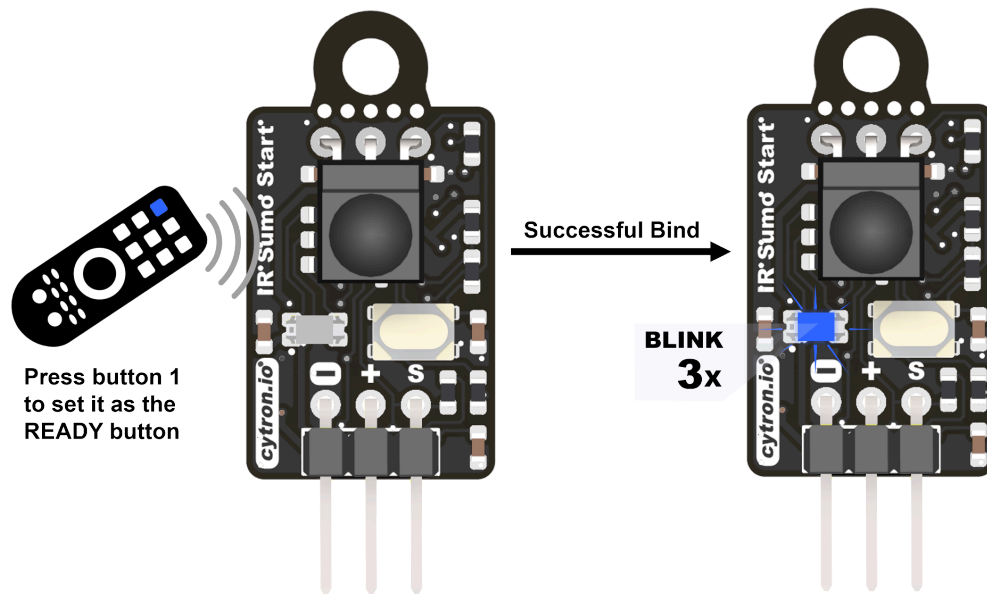


2. The LED should blink quickly in a sequence of **BLUE**, **GREEN**, and **RED** (100ms for each colour) for three times and then turn off. The module is now in Bind Mode and ready to accept new IR commands for binding.

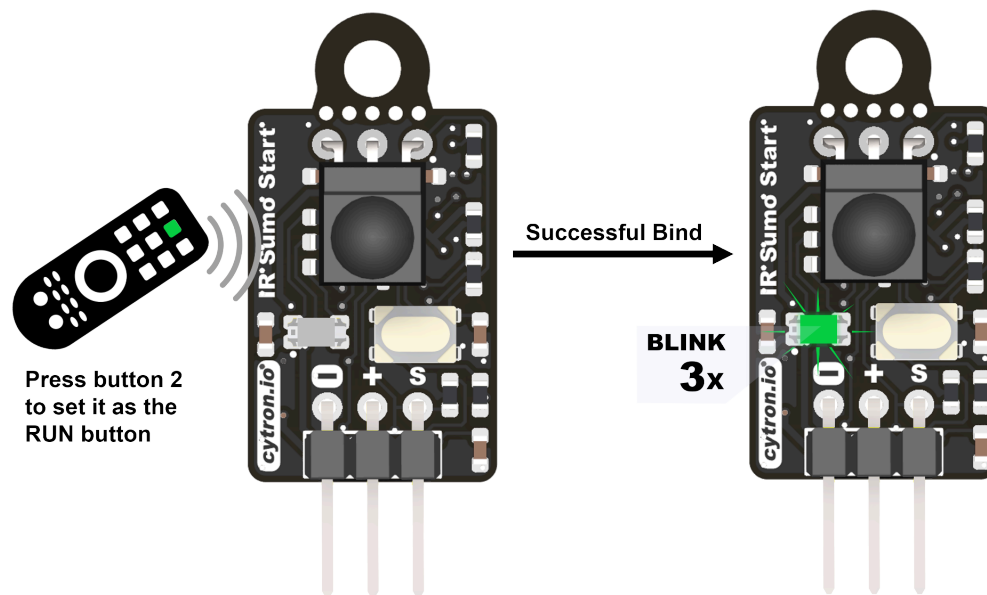


Bind new IR commands:

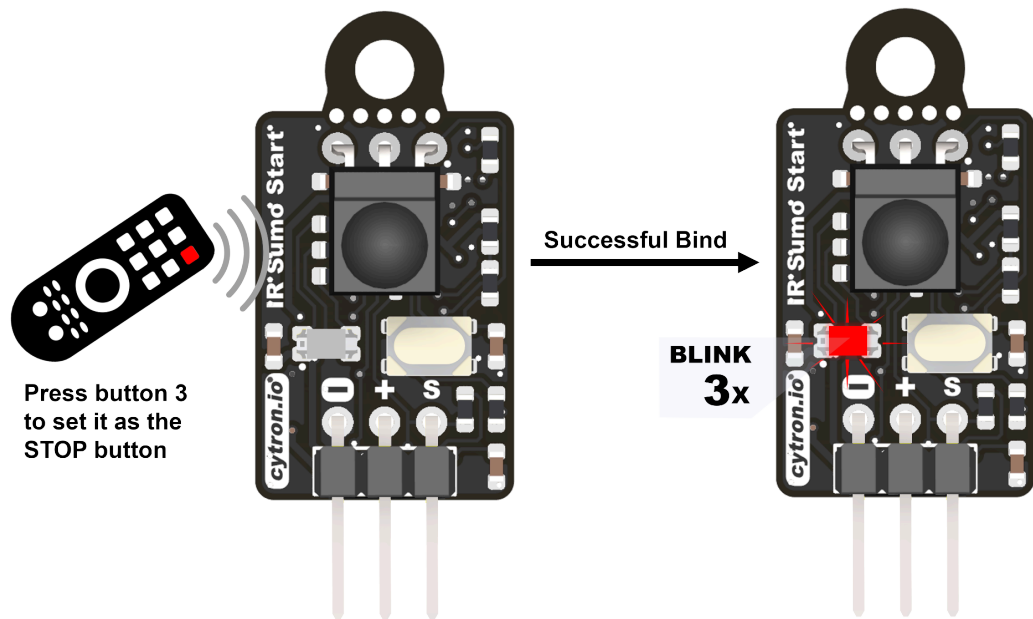
- Press button 1 of the IR remote to set it as the READY button. If binding is successful, the LED will blink BLUE three times then turn off.



- Press button 2 of the IR remote to set it as the RUN button. If binding is successful, the LED will blink GREEN three times then turn off.



- Press button 3 of the IR remote to set it as the STOP button. If binding is successful, the LED will blink RED three times then turn off. Now the module is automatically in Receive mode and ready to receive IR input.



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